

Hard Ops

The hub of the hard-surface workflow — booleans, bevels, mirroring, modifier management, and material/viewport helpers, all behind one menu.

WHEN Any destructive/boolean hard-surface work; managing the modifier stack.

main menu	Q
pie menu	Shift + Q
HOps Helper	Ctrl + `
materials	Alt + M
viewport	Alt + V
interactive mirror	Alt + X
bevel helpers	Ctrl + Shift + B

Booleans (object mode): Ctrl + Num - difference · + union · * intersect · ÷ slash.

BoxCutter

Draw-on-surface cutting. Sketch a shape and it becomes a live bevel inset, or new object. Pairs with Hard Ops.

WHEN Fast carving of panels, slots, and shapes directly in the viewport.

activate BoxCutter

Tool active: Ctrl + D pie menu · D helper · C custom.

MESHmachine

Post-bevel surgery: rework chamfers and fillets after the fact, fuse/unfuse/refuse surfaces, flatten, normal-edit, and stamp on detail with the Plug library.

WHEN Fixing or reshaping bevels/fillets that a normal bevel can't handle.

menu edit + object Y

All tools live inside this one menu.

Edge Flow

MENU-ONLY

Relaxes selected edges/loops to follow surrounding surface curve (like Maya's Edit Edge Flow). Also sets edge/vertex curve.

WHEN Cleaning up edge loops so they flow smoothly across curves.

Edge menu → Set Flow / Curve / Linear

Vertex menu → Set Vertex Curve

Also in the edit-mode Right-click context menu → Set Flow.

CURVEmachine

MESHmachine's little brother for curves — building and editing clean curve geometry for hard-surface detailing.

WHEN Cable runs, pipes, and curve-based detail work.

menu curve edit mode



Construction Lines

CAD-style accurate modelling — work to exact dimensions with construction guides, snapping, and numeric input.

WHEN Anything that needs real-world measurements and precise constraints.

launch key left of 1



Then follow the on-screen prompts.

Curves to Mesh

Generates a clean quad surface that flows between two or more

WHEN Building surfaces/patches from curve "rails."

run from the curve tools

While live: Alt + ↑↓↔ subdivisions · Ctrl + R refresh · Shift + Alt + Enter commit.

Grid Modeler

MENU-ONLY

Draw shapes on an interactive grid to cut into or build out geometry with precise grid snapping.

WHEN Greebles, panel detailing, and precise on-surface shapes.

Edit → Right-click → Grid Modeler

While live: Q / Enter confirm · Ctrl + Scroll resolution · Alt + S extend grid · W / RMB switch mode · Z close shape · X from edge.

MACHIN3tools

A large suite of quality-of-life tools and pie menus: Smart Vert/Edge/Face, Focus, Mirror, Group, customised pies, and Hard Ops / BoxCutter integration.

WHEN Everyday modelling speed-ups; ties the other tools together.

modes pie	Tab
focus	F / Ctrl + F
group	Ctrl + G
symmetrize	Alt + X
edge-constrained transform	Alt + R
smart select	Alt + LMB

Conform Object

MENU-ONLY

Wrap / conform one object onto the surface of another.

WHEN Laying detail, decals-as-geometry, or panels onto a curved surface.

Object → Object ▾ or Right-click → Conform Object

Quick Snap

Maya-style interactive vertex/point snapping — grab a point and precisely onto another.

WHEN Precise point-to-point placement during edit/object mode.

snap Ctrl +

Just Panels

Generates panelling/greeble detail, with an accompanying asset pack.

WHEN Fast sci-fi / hard-surface surface detailing.

pie menu Alt + K

Install the addon zip; load JustPanels_ASSETS via Preferences → File Paths → Asset Libraries.

